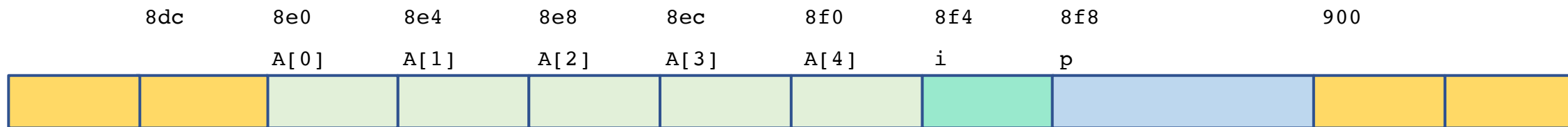




comp10002/comp20005 program pointer4.c

Support material for Chapter 7 of
Programming, Problem Solving, and Abstraction with C,
by Alistair Moffat

(c) 2023, The University of Melbourne
Prepared by Alistair Moffat, am Moffat@unimelb.edu.au



```
#define N 5
```

```
int
```

```
main(int argc, char *argv[]) {
```

```
    int A[N], i, *p;
```

```
    for (i=0; i<N; i++) {
```

```
        A[i] = i*(i+1)/2;
```

```
    }
```

```
    printf("A = %10p\n", A);
```

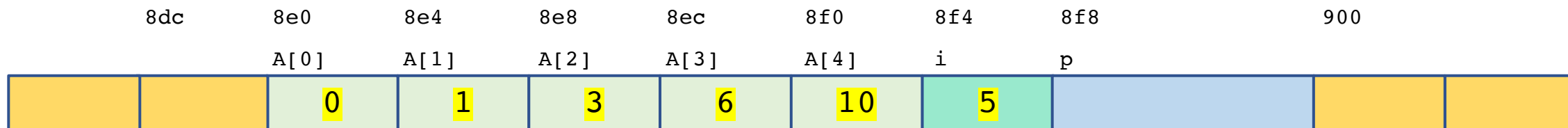
```
    for (p=A; p<A+N; p++) {
```

```
        printf("p = %10p, *p = %2d\n", p, *p);
```

```
    }
```

```
    return 0;
```

```
}
```



```
#define N 5
```

```
int
```

```
main(int argc, char *argv[]) {
```

```
    int A[N], i, *p;
```

```
    for (i=0; i<N; i++) {
```

```
        A[i] = i*(i+1)/2;
```

```
    }
```

```
    printf("A = %10p\n", A);
```

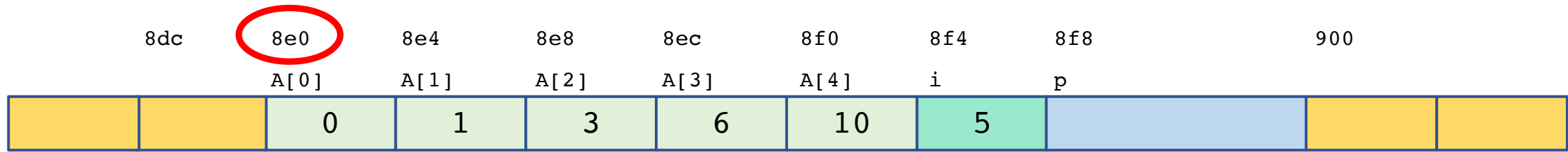
```
    for (p=A; p<A+N; p++) {
```

```
        printf("p = %10p, *p = %2d\n", p, *p);
```

```
    }
```

```
    return 0;
```

```
}
```



```
#define N 5
```

```
int
```

```
main(int argc, char *argv[]) {
```

```
    int A[N], i, *p;
```

```
    for (i=0; i<N; i++) {
```

```
        A[i] = i*(i+1)/2;
```

```
    }
```

```
    printf("A = %10p\n", A);
```

```
    for (p=A; p<A+N; p++) {
```

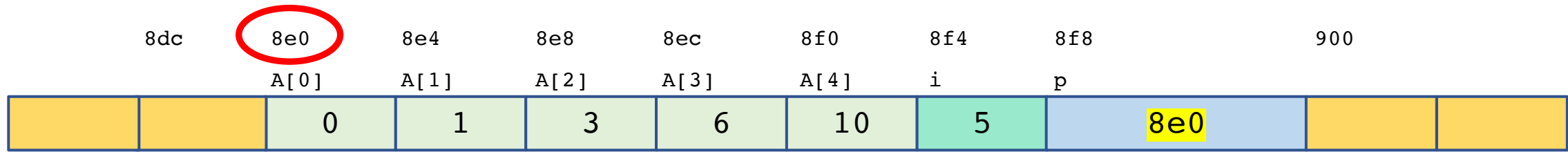
```
        printf("p = %10p, *p = %2d\n", p, *p);
```

```
    }
```

```
    return 0;
```

```
}
```

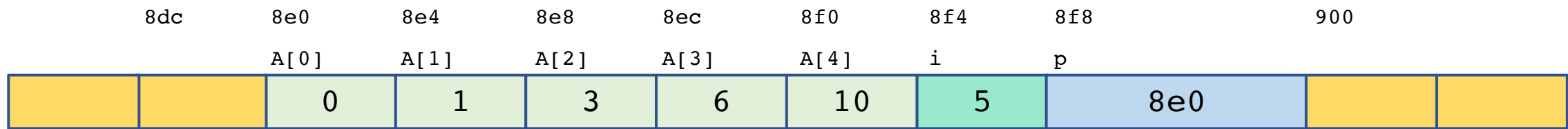
```
A = 0x7ffee353f8e0
```



```
#define N 5
```

```
int  
main(int argc, char *argv[]) {  
    int A[N], i, *p;  
  
    for (i=0; i<N; i++) {  
        A[i] = i*(i+1)/2;  
    }  
  
    printf("A = %10p\n", A);  
  
    for (p=A; p<A+N; p++) {  
        printf("p = %10p, *p = %2d\n", p, *p);  
    }  
  
    return 0;  
}
```

A = 0x7ffee353f8e0

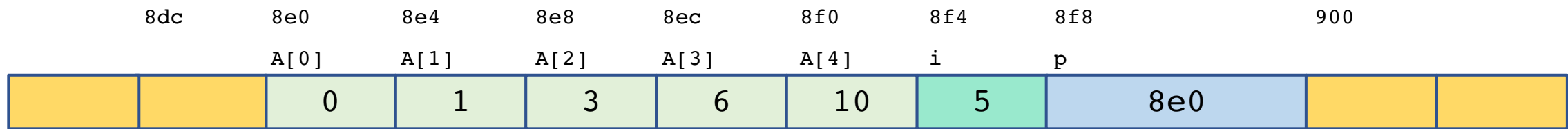


```
#define N 5
```

```
int  
main(int argc, char *argv[]) {  
    int A[N], i, *p;  
  
    for (i=0; i<N; i++) {  
        A[i] = i*(i+1)/2;  
    }  
  
    printf("A = %10p\n", A);  
  
    for (p=A; p<A+N; p++) {  
        printf("p = %10p, *p = %2d\n", p, *p);  
    }  
  
    return 0;  
}
```

A = 0x7ffee353f8e0

A+N = 8e0 + 5 x int = 8f4

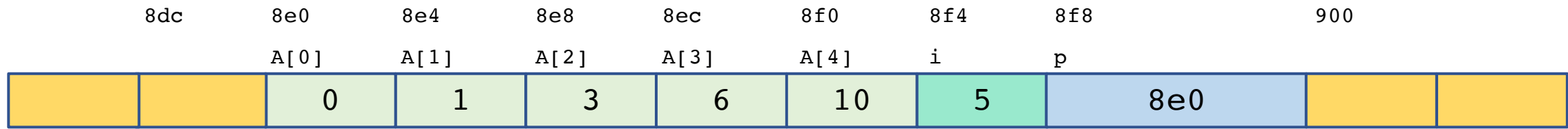


```
#define N 5
```

```
int  
main(int argc, char *argv[]) {  
    int A[N], i, *p;  
  
    for (i=0; i<N; i++) {  
        A[i] = i*(i+1)/2;  
    }  
  
    printf("A = %10p\n", A);  
  
    for (p=A; p<A+N; p++) {  
        printf("p = %10p, *p = %2d\n", p, *p);  
    }  
  
    return 0;  
}
```

```
A = 0x7ffee353f8e0
```

```
p = 0x7ffee353f8e0, *p = 0
```



```
#define N 5
```

```
int
main(int argc, char *argv[]) {
    int A[N], i, *p;

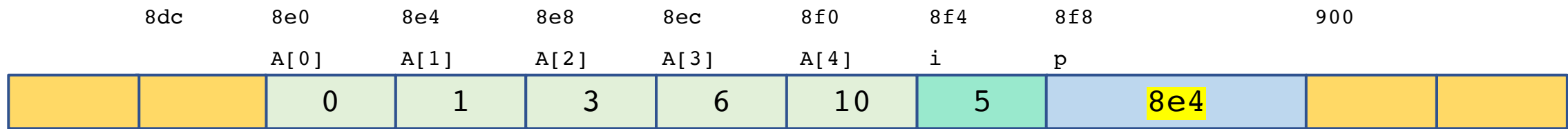
    for (i=0; i<N; i++) {
        A[i] = i*(i+1)/2;
    }
    p+1 = 8e0 + 1 x int = 8e4

    printf("A = %10p\n", A);

    for (p=A; p<A+N; p++) {
        printf("p = %10p, *p = %2d\n", p, *p);
    }

    return 0;
}
```

```
A = 0x7ffee353f8e0
p = 0x7ffee353f8e0, *p = 0
```

```
#define N 5
```

```
int
main(int argc, char *argv[]) {
    int A[N], i, *p;

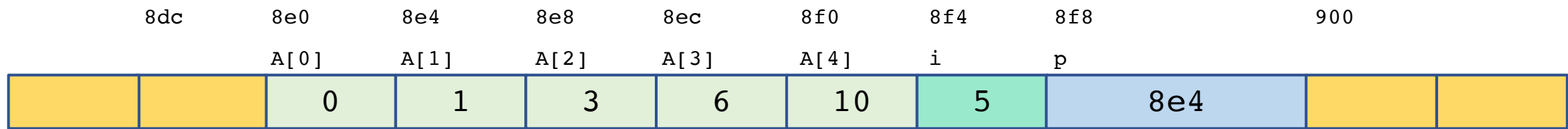
    for (i=0; i<N; i++) {
        A[i] = i*(i+1)/2;
    }

    printf("A = %10p\n", A);

    for (p=A; p<A+N; p++) {
        printf("p = %10p, *p = %2d\n", p, *p);
    }

    return 0;
}
```

```
A = 0x7ffee353f8e0
p = 0x7ffee353f8e0, *p = 0
```



```
#define N 5
```

```
int
main(int argc, char *argv[]) {
    int A[N], i, *p;

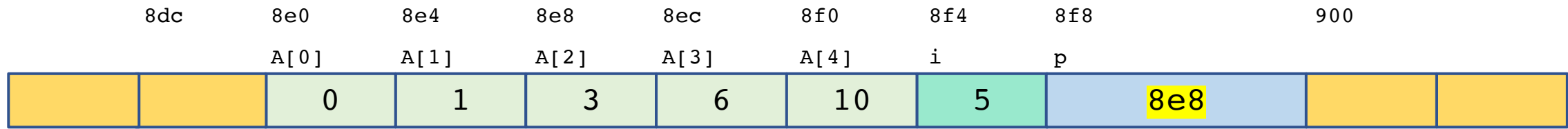
    for (i=0; i<N; i++) {
        A[i] = i*(i+1)/2;
    }

    printf("A = %10p\n", A);

    for (p=A; p<A+N; p++) {
        printf("p = %10p, *p = %2d\n", p, *p);
    }

    return 0;
}
```

```
A = 0x7ffee353f8e0
p = 0x7ffee353f8e0, *p = 0
p = 0x7ffee353f8e4, *p = 1
```



```
#define N 5
```

```
int
main(int argc, char *argv[]) {
    int A[N], i, *p;

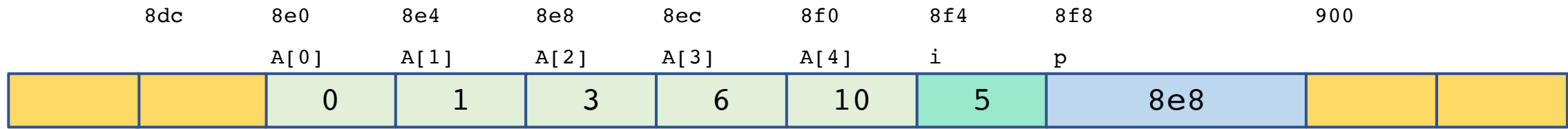
    for (i=0; i<N; i++) {
        A[i] = i*(i+1)/2;
    }

    printf("A = %10p\n", A);

    for (p=A; p<A+N; p++) {
        printf("p = %10p, *p = %2d\n", p, *p);
    }

    return 0;
}
```

```
A = 0x7ffee353f8e0
p = 0x7ffee353f8e0, *p = 0
p = 0x7ffee353f8e4, *p = 1
```



```
#define N 5
```

```
int
main(int argc, char *argv[]) {
    int A[N], i, *p;

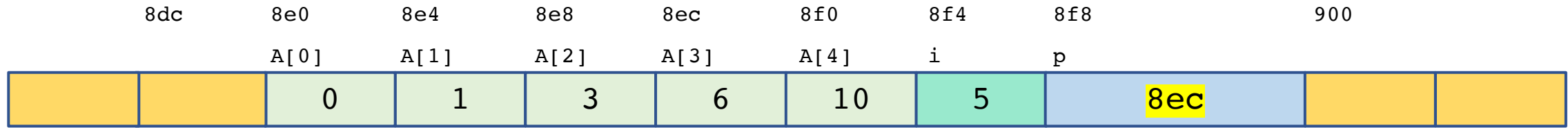
    for (i=0; i<N; i++) {
        A[i] = i*(i+1)/2;
    }

    printf("A = %10p\n", A);

    for (p=A; p<A+N; p++) {
        printf("p = %10p, *p = %2d\n", p, *p);
    }

    return 0;
}
```

```
A = 0x7ffee353f8e0
p = 0x7ffee353f8e0, *p = 0
p = 0x7ffee353f8e4, *p = 1
p = 0x7ffee353f8e8, *p = 3
```



```
#define N 5
```

```
int
main(int argc, char *argv[]) {
    int A[N], i, *p;

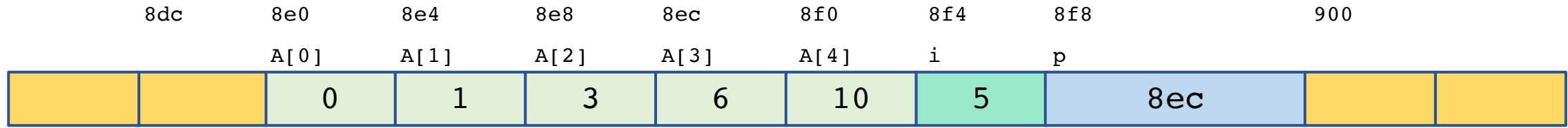
    for (i=0; i<N; i++) {
        A[i] = i*(i+1)/2;
    }

    printf("A = %10p\n", A);

    for (p=A; p<A+N; p++) {
        printf("p = %10p, *p = %2d\n", p, *p);
    }

    return 0;
}
```

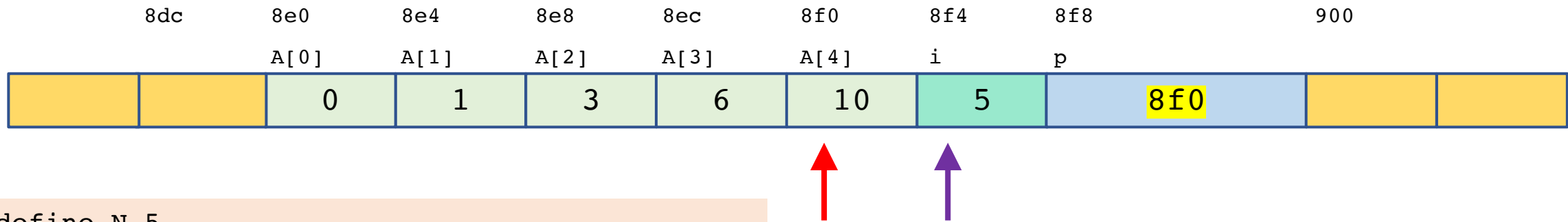
```
A = 0x7ffee353f8e0
p = 0x7ffee353f8e0, *p = 0
p = 0x7ffee353f8e4, *p = 1
p = 0x7ffee353f8e8, *p = 3
```



```
#define N 5
```

```
int  
main(int argc, char *argv[]) {  
    int A[N], i, *p;  
  
    for (i=0; i<N; i++) {  
        A[i] = i*(i+1)/2;  
    }  
  
    printf("A = %10p\n", A);  
  
    for (p=A; p<A+N; p++) {  
        printf("p = %10p, *p = %2d\n", p, *p);  
    }  
  
    return 0;  
}
```

```
A = 0x7ffee353f8e0  
p = 0x7ffee353f8e0, *p = 0  
p = 0x7ffee353f8e4, *p = 1  
p = 0x7ffee353f8e8, *p = 3  
p = 0x7ffee353f8ec, *p = 6
```



```
#define N 5
```

```
int
main(int argc, char *argv[]) {
    int A[N], i, *p;

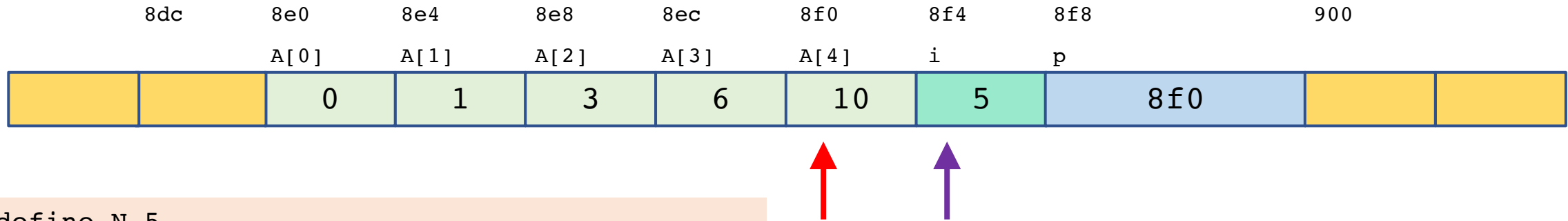
    for (i=0; i<N; i++) {
        A[i] = i*(i+1)/2;
    }

    printf("A = %10p\n", A);

    for (p=A; p<A+N; p++) {
        printf("p = %10p, *p = %2d\n", p, *p);
    }

    return 0;
}
```

```
A = 0x7ffee353f8e0
p = 0x7ffee353f8e0, *p = 0
p = 0x7ffee353f8e4, *p = 1
p = 0x7ffee353f8e8, *p = 3
p = 0x7ffee353f8ec, *p = 6
```



```
#define N 5
```

```
int
main(int argc, char *argv[]) {
    int A[N], i, *p;

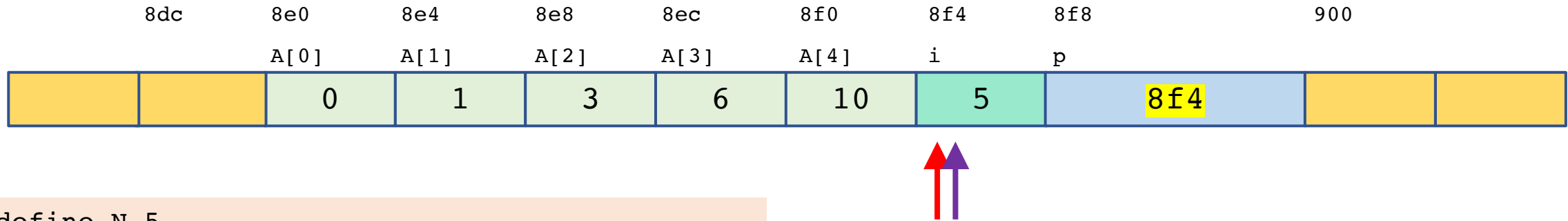
    for (i=0; i<N; i++) {
        A[i] = i*(i+1)/2;
    }

    printf("A = %10p\n", A);

    for (p=A; p<A+N; p++) {
        printf("p = %10p, *p = %2d\n", p, *p);
    }

    return 0;
}
```

```
A = 0x7ffee353f8e0
p = 0x7ffee353f8e0, *p = 0
p = 0x7ffee353f8e4, *p = 1
p = 0x7ffee353f8e8, *p = 3
p = 0x7ffee353f8ec, *p = 6
p = 0x7ffee353f8f0, *p = 10
```

```
#define N 5
```

```
int
main(int argc, char *argv[]) {
    int A[N], i, *p;

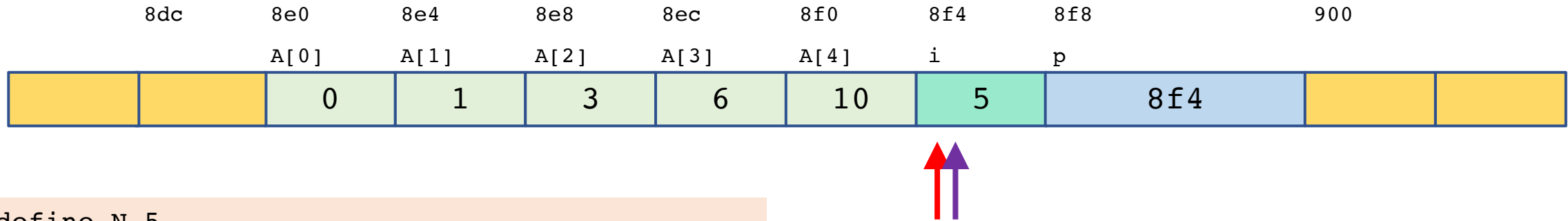
    for (i=0; i<N; i++) {
        A[i] = i*(i+1)/2;
    }

    printf("A = %10p\n", A);

    for (p=A; p<A+N; p++) {
        printf("p = %10p, *p = %2d\n", p, *p);
    }

    return 0;
}
```

```
A = 0x7ffee353f8e0
p = 0x7ffee353f8e0, *p = 0
p = 0x7ffee353f8e4, *p = 1
p = 0x7ffee353f8e8, *p = 3
p = 0x7ffee353f8ec, *p = 6
p = 0x7ffee353f8f0, *p = 10
```



```
#define N 5
```

```
int
main(int argc, char *argv[]) {
    int A[N], i, *p;

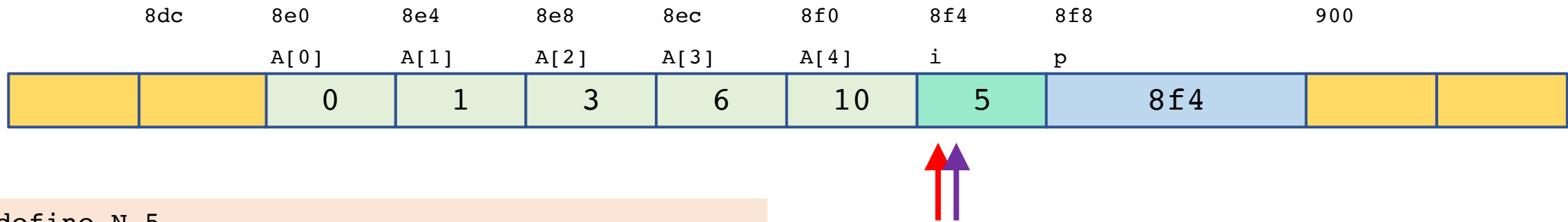
    for (i=0; i<N; i++) {
        A[i] = i*(i+1)/2;
    }

    printf("A = %10p\n", A);

    for (p=A; p<A+N; p++) {
        printf("p = %10p, *p = %2d\n", p, *p);
    }

    return 0;
}
```

```
A = 0x7ffee353f8e0
p = 0x7ffee353f8e0, *p = 0
p = 0x7ffee353f8e4, *p = 1
p = 0x7ffee353f8e8, *p = 3
p = 0x7ffee353f8ec, *p = 6
p = 0x7ffee353f8f0, *p = 10
```



```
#define N 5
```

```
int
main(int argc, char *argv[]) {
    int A[N], i, *p;

    for (i=0; i<N; i++) {
        A[i] = i*(i+1)/2;
    }

    printf("A = %10p\n", A);

    for (p=A; p<A+N; p++) {
        printf("p = %10p, *p = %d\n", p, *p);
    }

    return 0;
}
```

```
A = 0x7ffee353f8e0
p = 0x7ffee353f8e0, *p = 0
p = 0x7ffee353f8e4, *p = 1
p = 0x7ffee353f8e8, *p = 3
p = 0x7ffee353f8ec, *p = 6
p = 0x7ffee353f8f0, *p = 10
```

	8dc	8e0	8e4	8e8	8ec	8f0	8f4	8f8		900
		A[0]	A[1]	A[2]	A[3]	A[4]	i	p		
		0	1	3	6	10	5	8f4		

```
#define N 5
```

```
int
main(int argc, char *argv[]) {
    int A[N], i, *p;

    for (i=0; i<N; i++) {
        A[i] = i*(i+1)/2;
    }

    printf("A = %10p\n", A);

    for (p=A; p<A+N; p++) {
        printf("p = %10p, *p = %2d\n", p, *p);
    }

    return 0;
}
```

```
A = 0x7ffee353f8e0
p = 0x7ffee353f8e0, *p = 0
p = 0x7ffee353f8e4, *p = 1
p = 0x7ffee353f8e8, *p = 3
p = 0x7ffee353f8ec, *p = 6
p = 0x7ffee353f8f0, *p = 10
```

	8dc	8e0	8e4	8e8	8ec	8f0	8f4	8f8		900
		A[0]	A[1]	A[2]	A[3]	A[4]	i	p		
		0	1	3	6	10	5	8f4		

```
#define N 5
```

```
int
main(int argc, char *argv[]) {
    int A[N], i, *p;

    for (i=0; i<N; i++) {
        A[i] = i*(i+1)/2;
    }

    printf("A = %10p\n", A);

    for (p=A; p<A+N; p++) {
        printf("p = %10p, *p = %2d\n", p, *p);
    }

    return 0;
}
```

```
A = 0x7ffee353f8e0
p = 0x7ffee353f8e0, *p = 0
p = 0x7ffee353f8e4, *p = 1
p = 0x7ffee353f8e8, *p = 3
p = 0x7ffee353f8ec, *p = 6
p = 0x7ffee353f8f0, *p = 10
mac:
```